RhythmRoute: Trajectory-Conditioned Dance Generation from Music

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***Abstract*—State-of-the-art dance generation models like EDGE excel at creating realistic, music-driven choreography but lack direct spatial control, limiting their use in navigable environments. We address this by introducing *RhythmRoute*, a framework that integrates precise path-following capabilities into a dance diffusion model. Our method implements a core Transformer architecture to accept dual conditions: it processes a 3D trajectory via a dedicated encoder while simultaneously leveraging rich Jukebox music features. By incorporating a trajectory loss into the training objective, our model learns to generate motions that are both stylistically faithful to the music and spatially anchored to the user-defined path. To quantify this path adherence, we introduce the Mean Trajectory Error (MTE) metric. *RhythmRoute* produces physically plausible dances that adhere to user-defined trajectories, offering an intuitive and powerful control modality for interactive entertainment, gaming, and virtual choreography.**

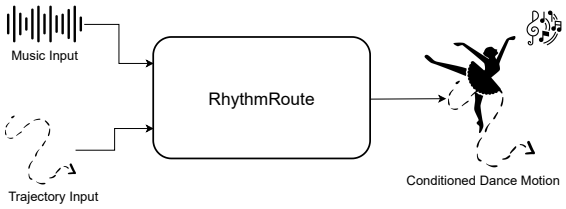


Fig. 1. Conceptual overview of the RhythmRoute framework. Our model takes two distinct conditioning inputs: a high-level music signal for stylistic and rhythmic guidance, and a low-level 3D root trajectory for precise spatial control. The RhythmRoute model processes these inputs simultaneously to generate a high-fidelity, full-body dance sequence. The resulting motion is a synthesis that is both musically synchronized and spatially constrained to the user-defined path.

I. Introduction

The synthesis of human motion from data has seen remarkable progress, with the generation of dance from musical input standing out as a notable achievement. Modern generative systems can produce complex choreographies that capture the rhythm, tempo, and mood of an audio track with high fidelity. These models have demonstrated an impressive ability to learn the implicit connection between acoustic patterns and the dynamics of human movement.

However, a fundamental gap exists in their capabilities: a lack of directorial control. These systems function as unconstrained generators, producing motion without providing a mechanism for users to guide a character's location or path through a scene. The dancer's trajectory is an emergent outcome of the creative process rather than a controllable parameter. This lack of navigational autonomy presents a significant barrier to their adoption in interactive contexts such as video games, virtual reality, and simulated environments, where characters must adhere to specific paths or interact with their surroundings.

This thesis introduces RhythmRoute, a framework designed to bridge this gap by incorporating explicit spatial control into a music-driven dance generation model. We present a system that conditions the synthesis process on two parallel inputs: the stylistic information from a piece of music and the navigational constraints of a user-defined 3D trajectory. Our approach implements a diffusion-based architecture to process both modalities, enabling it to produce motion that is simultaneously responsive to the music and compliant with the specified path.

To accomplish this, we introduce a dedicated trajectory encoding module to interpret the user-defined path. We augment the learning process with a new trajectory loss term, which penalizes any deviation between the generated motion and the target path. This technical contribution ensures that the model learns not just how to dance, but how to adapt its movements to follow a specific route. To measure this new capability, we also propose a Mean Trajectory Error (MTE) metric. Our evaluation confirms that RhythmRoute can generate physically coherent dance sequences that follow user-provided trajectories while maintaining musical synchrony, thereby offering a new level of control for creative and interactive applications.

Our contributions are as follows:

- A novel framework for conditioning a dance diffusion model on both high-level musical features and a low-level 3D trajectory.
- A modified Transformer architecture that uses a dedicated Trajectory Encoder and cross-attention to integrate dual-modality inputs.
- A new training objective that incorporates a trajectory loss alongside physical plausibility losses to enable spatial control without compromising motion quality.
- A new quantitative evaluation protocol for such systems, using a Trajectory Error metric to measure control accuracy alongside existing metrics for motion quality.

II. Related Work

Our work lies at the intersection of two primary domains in human motion synthesis: music-conditioned dance generation and trajectory-conditioned motion control. We aim to combine the strengths of both paradigms.

A. Human Motion Generation

The automated synthesis of human motion from data is a foundational problem in computer graphics. The field has progressed from early procedural and statistical methods towards deep learning-based approaches. Generative models, including Generative Adversarial Networks (GANs) [Goodfellow et al. 2014], Variational Autoencoders (VAEs) [Kingma and Welling 2014], and more recently, Denoising Diffusion Probabilistic Models (DDPMs) [Ho, Jain, and Abbeel 2020], have become central to generating diverse and realistic motion sequences. These technologies provide the architectural backbone for the specialized tasks of dance generation and motion control.

B. Diffusion Models for Motion Generation

Denoising Diffusion Probabilistic Models (DDPMs), first proposed Ho, Jain, and Abbeel 2020, have emerged as an exceptionally powerful class of generative models. They operate by progressively corrupting data with Gaussian noise and then training a neural network to reverse this process. This iterative denoising procedure allows the model to synthesize highly detailed and complex data distributions, making it ideal for generating human motion. The core of these models is typically a U-Net [Ronneberger, Fischer, and Brox 2015] or Transformer [Vaswani et al. 2017] architecture that refines a noisy input over many timesteps, often conditioned on external data such as text or audio embeddings. Our research leverages the power of this framework as the generative backbone for our system.

C. Music-Conditioned Dance Generation

The task of generating realistic and rhythmically aligned dance from music has seen significant progress. Early methods often relied on regressive models [Tang, Jia, and Mao 2018, Ruilong Li et al. 2021, Le et al. 2023, B. Li et al. 2022] or generative adversarial networks [Lee et al. 2019, Kim et al. 2022] to map audio features to motion sequences. However, recent state-of-the-art results have been achieved using diffusion models, which excel at generating high-fidelity, diverse outputs.

The FineDance [Ronghui Li, Zhao, et al. 2023] paper introduces FineNet, which uses a diffusion model with separate expert networks for body and hand motions. This specialized approach allows the network to generate more realistic and expressive full-body dance sequences. Lodge [Ronghui Li, Y. Zhang, et al. 2024], on the other hand, proposes a network designed to generate long and expressive dance sequences from music. Its model uses a two-stage, coarse-to-fine diffusion process. The first stage, global diffusion, creates key dance movements, and the second, local diffusion, generates the detailed motions, guided by these key movements.

Our work builds upon EDGE: Editable Dance Generation from Music [Tseng, Castellon, and Liu 2023], another powerful transformer-based diffusion model for music-to-dance synthesis. The key contributions of EDGE are threefold: (1) the use of a deep Transformer architecture that operates directly on a rich, 151-dimensional motion representation, (2) the leveraging of Jukebox [Dhariwal et al. 2020], a powerful pre-trained music model, as a feature extractor to provide a nuanced conditioning signal, and (3) a suite of auxiliary loss functions, including forward kinematics and foot contact consistency losses, that ensure the generated motion is physically plausible and free of common artifacts like foot-skating.

However, the primary motivation for our work is a key limitation in existing models like EDGE and FineDance. These systems focus on the quality and musicality of the generated poses but lack mechanisms for explicit global trajectory control. They do not provide a method to direct a character along a user-defined path. The model generates its own trajectory based on the music's style rather than conforming to an external spatial constraint.

D. Trajectory-Conditioned Motion Synthesis

A parallel and highly active area of research focuses on controlling motion using textual descriptions and spatial constraints. Models like MDM [Tevet et al. 2023] and MotionDiffuse [M. Zhang et al. 2024] demonstrated that diffusion models could generate high-quality motion from text prompts alone.

More recently, frameworks have emerged that combine both language and trajectory control. Guided Motion Diffusion (GMD) [Karunratanakul et al. 2023] presents a method to control text-based human motion generation with spatial constraints like trajectories and keyframes. The model architecture uses a two-stage diffusion process, first for the trajectory and then for the full-body motion. Whereas, OmniControl [Xie et al. 2024] introduces a method for text-based human motion generation that allows for precise spatial control over any joint at any time. The model architecture enhances a standard motion diffusion model with a novel hybrid guidance module.

Our work is heavily inspired by TLControl [Wan et al. 2024], a state-of-the-art model in this domain. TLControl introduces a sophisticated two-stage process: first, it learns a compact and structured "motion vocabulary" using a part-based VQ-VAE, which disentangles the movement of different body parts. Second, it uses a Masked Trajectory Transformer to predict a sequence of these motion tokens based on a text prompt and a partial user-drawn trajectory. A key contribution of TLControl is its use of test-time optimization, which refines the initial motion prediction to ensure it precisely adheres to the specified trajectory constraints. This provides a high degree of accuracy and flexibility.

Despite their strengths, these models are designed for general-purpose actions like "walking" or "jumping." They have no intrinsic understanding of rhythm, beat alignment, or the complex, non-functional movements that define dance. A model like TLControl, GMD, or Omnicontrol might execute a

path accurately, but it would lack the musical expression that is the core requirement of our task. Our work aims to address this by creating a system that does not sacrifice musicality for control.

III. Methodology

A. Motion and Music Representation

Our model operates on a 151-dimensional feature vector for each frame, processed at 30 FPS. This representation is derived from the Skinned Multi-Person Linear (SMPL) model format [Loper et al. 2015], which provides both per-joint rotations and the root's global translation. We transform this raw data into a feature vector containing three key components: the global root position (3 dimensions), the local rotation of all 24 body joints in a continuous 6D format (144 dimensions), and binary foot contact flags (4 dimensions). The availability of the explicit root trajectory within the SMPL data makes it uniquely suited for our task. For training, this entire 151-dimensional vector is normalized to the range [-1, 1].

To provide a rich conditioning signal, we use OpenAI's Jukebox model [Dhariwal et al. 2020] as a music feature extractor. Following EDGE, we extract activations from an intermediate layer of the Jukebox encoder. This converts each 5-second audio clip into a sequence of 4800-dimensional feature vectors at 30 FPS. This embedding captures musical concepts like rhythm and timbre and serves as the stylistic guidance for the generation process.

B. Diffusion Model Framework

Our approach is built upon the foundation of DDPMs [Ho, Jain, and Abbeel 2020]. The core mechanism of a DDPM consists of two processes: a fixed forward noising process and a learned reverse denoising process.

1) Forward Process: The forward process, q, gradually adds Gaussian noise to the clean data $\mathbf{x}_0$ over a series of T discrete timesteps. The data at any timestep t is given by a closed-form equation:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \tag{1}$$

where $\bar{\alpha}_t$ is a pre-defined noise schedule that controls the signal-to-noise ratio at each step. As t approaches T, $\bar{\alpha}_t$ approaches 0, and $\mathbf{x}_T$ becomes indistinguishable from pure Gaussian noise.

2) Reverse Process: The generative power of the model comes from the learned reverse process, p_θ, which aims to reverse this noising operation. The model learns to iteratively denoise the data, starting from pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and predicting a slightly cleaner sample $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$ at each step. This process is guided by a neural network, $\hat{\mathbf{x}}_\theta$, which is trained to predict the original clean data $\mathbf{x}_0$ from a given noisy sample $\mathbf{x}_t$ and timestep t. The training objective, often referred to as $\mathcal{L}_{\text{simple}}$, is typically a Mean Squared Error (MSE) between the ground truth and the model's prediction:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0,t}\left[\|\mathbf{x}_0 - \hat{\mathbf{x}}_\theta(\mathbf{x}_t, t, c)\|_2^2\right], \tag{2}$$

where c is the input conditions.

In our work, the neural network $\hat{\mathbf{x}}_\theta$ is a Transformer-based architecture conditioned not only on the timestep t, but also on external guidance signals of music and trajectories, as detailed in the following sections. The reverse process is performed using a Denoising Diffusion Implicit Model (DDIM) sampler [Song, Meng, and Ermon 2021], which allows for faster and more efficient generation by skipping timesteps.

C. Trajectory Processing and Encoding

The model's ability to follow a specified path is contingent on trajectory processing and encoding pipeline. This pipeline transforms a raw sequence of 3D coordinates into a rich, latent representation that guides the motion generation process. It involves normalization, a data augmentation strategy through masking, and a multi-stage Transformer-based encoding architecture, Fig. 2(a).

We chose a Transformer for the trajectory encoder due to its proven effectiveness in capturing long-range temporal dependencies in sequence data, allowing it to understand the global shape of the path rather than just local points.

1) Input Trajectory and Normalization: The trajectory input is a sequence of 3D root positions ($\mathbf{P} \in \mathbb{R}^{B\times S\times 3}$), where B is the batch size and $S = 150$ is the sequence length. This serves as both a conditioning input and the ground truth for our trajectory loss. For our primary model, we normalize ($\mathbf{P}$) using the same Min-Max scaling as the main 151-dimensional motion vector, ensuring consistent data processing. As part of our ablation studies (Sec. IV C), we also evaluate the performance of Z-score normalization, where the trajectory is scaled using a separately computed mean (μ_{traj}) and standard deviation (σ_{traj}):

$$\mathbf{P}\text{norm} = \frac{\mathbf{P} - \mu_{\text{traj}}}{\sigma_{\text{traj}}} \tag{3}$$

Unless otherwise stated, all results use Min-Max scaling.

2) Trajectory Masking: To improve model robustness and prevent over-reliance on the trajectory, a masking-based data augmentation strategy is employed. During training, with a 25% probability, random contiguous segments of the input trajectory are masked (i.e., zeroed out). This forces the model to learn to generate plausible motion sequences by relying on the musical context, effectively in-painting the missing trajectory information.

3) Trajectory Encoding: The (potentially masked) normalized trajectory, ($\mathbf{P}_{\text{masked}}$), is then fed into a dedicated Trajectory Encoder. This encoder is a Transformer-based architecture designed to convert the coordinate sequence into a rich latent representation. The process consists of three stages:

- **Input Projection**: A linear layer first projects the 3D input coordinates into the model's high-dimensional latent space ($D = 512D$).
- **Positional Encoding**: Standard sinusoidal positional encodings are added to the input projection to provide the model with explicit information about the temporal order of the trajectory points.
- **Transformer Encoder Stack**: The resulting sequence is processed by a stack of 4 standard Transformer encoder

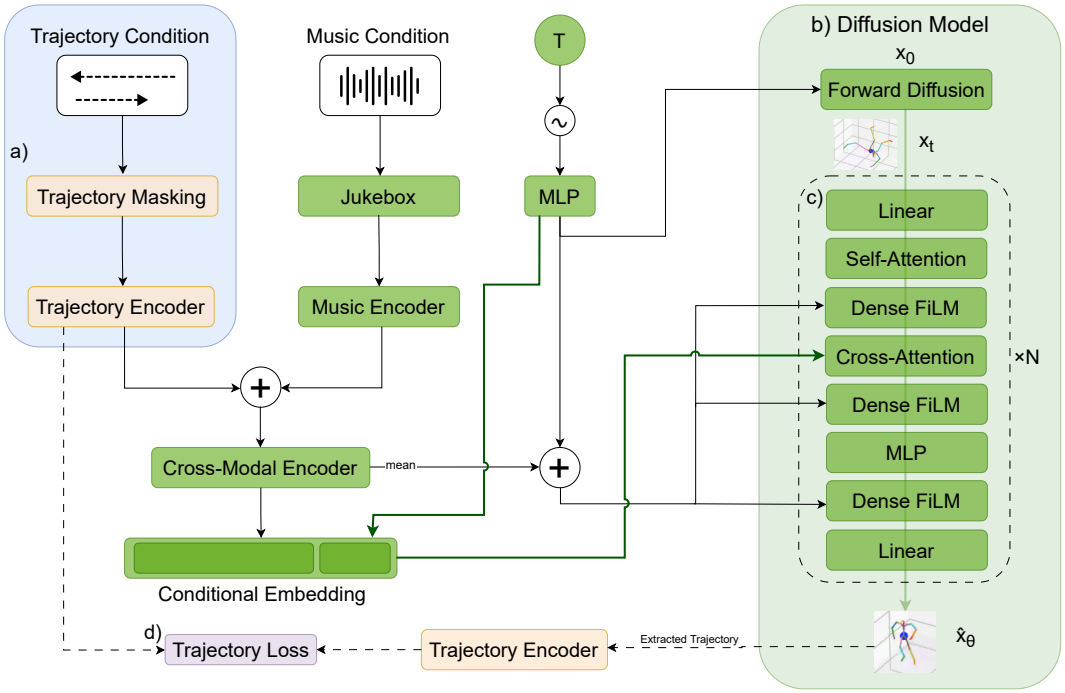


Fig. 2. **RhythmRoute** Pipeline Overview: a) Trajectory Encoder Module, b) Diffusion Model, c) Transformer Decoder, and d) Trajectory Loss Function - During training, the trajectory is extracted from the dance pose predicted by the diffusion model at a given timestep. This predicted trajectory is then passed through the Trajectory Encoder to create a feature vector. In parallel, the ground-truth trajectory is passed through the same Trajectory Encoder. The final loss is the Mean Squared Error between these two resulting encoded representations.

layers. Each layer uses self-attention to refine the trajectory representation. The output is a sequence of latent embeddings, ($\mathbf{c}_{\text{traj}}$), that captures the spatial and temporal characteristics of the conditioning trajectory.

4) Fusion with Music Conditioning: Finally, this encoded trajectory ($\mathbf{c}_{\text{traj}}$) is fused with the separately encoded music embeddings, ($\mathbf{c}_{\text{music}}$), to create a unified, multimodal context. The trajectory and music embeddings are concatenated along the sequence dimension and then passed through a linear projection layer, which reduces the sequence length back to S. This concatenate-and-project fusion strategy forces the model to learn a compact and meaningful joint representation of the two modalities rather than treating them as independent signals. This fused representation is then passed through a final two-layer Transformer encoder, producing the final conditioning tensor, ($\mathbf{c} = \mathbf{c}_{\text{music}} + \mathbf{c}_{\text{traj}}$), which is supplied to the cross-attention mechanisms of the main `DanceDecoder`.

D. Transformer Decoder

The core of our generative model, $\hat{\mathbf{x}}_\theta$, is a Transformer Decoder architecture, which is referred to as `DanceDecoder`, Fig. 2(c). Transformer, introduced in Vaswani et al. 2017, is well-suited for sequential data like human motion due to its powerful attention mechanism. The `DanceDecoder` is responsible for learning the complex temporal and multimodal relationships between the evolving motion, the guiding music, and the target trajectory.

The architecture is composed of a stack of 8 identical Transformer layers. During a single denoising step, each layer processes a 150-frame sequence of the current noisy motion representation, x_t, refining it based on the conditioning inputs. The data flow through each layer involves four key operations:

1) Self-Attention: This is the foundational mechanism of the Transformer. For each frame in the motion sequence, self-attention dynamically computes weights to determine which other frames in the 150-frame window are most relevant for predicting the current pose. This allows the model to capture long-range temporal dependencies and ensure the entire motion is coherent.

2) Multimodal Cross-Attention: Within each layer of the Transformer decoder, a single cross-attention mechanism is employed to integrate the conditional guidance. The motion representation, which is the output of the self-attention layer, acts as the query. This query attends to the fused multimodal condition embedding ($\mathbf{c}_{\text{music}} + \mathbf{c}_{\text{traj}}$), which serves as both the key and value. This unified attention step allows the model to simultaneously synthesize stylistic information from the music and spatial constraints from the trajectory, ensuring the generated pose is contextually appropriate to both inputs at every step of the denoising process.

3) Feed-Forward Network with FiLM: Finally, the output is passed through a standard feed-forward network for further processing. We also incorporate the diffusion timestep, t, at this stage using Feature-wise Linear Modulation (FiLM) [Perez et al. 2018]. The timestep is first converted to an

embedding, which is then used to generate a scale and shift parameter. These parameters dynamically modulate the output of the feed-forward network, allowing the model to adapt its behavior based on the noise level of the input.

E. Auxiliary Loss Functions

Our final training objective is a composite loss function that balances our novel trajectory objective with the established plausibility objectives from the original EDGE model [Tseng, Castellon, and Liu 2023]. In addition to the primary reconstruction loss ($\mathcal{L}_{\text{simple}}$, Eq. (2)), we adopt the velocity, forward kinematics, and foot contact consistency losses. To this, we add our novel trajectory loss, which we detail below.

1) Velocity Loss: To ensure temporal smoothness and prevent jerky movements, we use a velocity loss. This loss compares the frame-to-frame velocity of the predicted motion sequence to that of the ground truth sequence. N denotes the total number of frames and i denotes the frame index.

$$\mathcal{L}_{\text{vel}} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|(\mathbf{x}_0^{(i+1)} - \mathbf{x}_0^{(i)}) - (\hat{\mathbf{x}}_\theta^{(i+1)} - \hat{\mathbf{x}}_\theta^{(i)})\|_2^2 \quad (4)$$

2) Forward Kinematics / Joint Loss: To enforce physical plausibility of the entire skeleton, we use a forward kinematics loss. The predicted and ground truth rotation parameters are converted into 3D world-space joint positions via the SMPL forward kinematics function, FK($\cdot$). The loss is then computed on the resulting 3D skeletons. This directly penalizes physically impossible poses.

$$\mathcal{L}_{\text{joint}} = \frac{1}{N} \sum_{i=1}^{N} \|\text{FK}(\mathbf{x}_0^{(i)}) - \text{FK}(\hat{\mathbf{x}}_\theta^{(i)})\|_2^2 \quad (5)$$

3) Contact Consistency Loss: To prevent the common "foot skate" artifact, we employ a contact consistency loss. This loss penalizes the model if it predicts that a foot is in contact with the ground while also predicting that the foot is moving. $\hat{b}$ represents the model's own prediction of the binary foot contact label.

$$\mathcal{L}_{\text{contact}} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|(\text{FK}(\hat{\mathbf{x}}_\theta^{(i+1)}) - \text{FK}(\hat{\mathbf{x}}_\theta^{(i)})) \cdot \hat{\mathbf{b}}^{(i)}\|_2^2 \quad (6)$$

4) Trajectory Loss: This is our primary contribution to the objective function. To ensure the model strictly adheres to the user-defined path, we introduce a dedicated trajectory loss. After the model produces its prediction of the full 151-dimensional vector, we extract the predicted root position, $\hat{x}_{root}$. We then compute the MSE between the encoded predicted path and the encoded ground truth trajectory, x_{root}, that was provided as an input condition. The $\mathcal{E}_{traj}(\cdot)$ denotes the *Trajectory Encoder*, Fig. 2(d).

$$\mathcal{L}_{traj} = \|\mathcal{E}_{traj}(\hat{x}_{root}) - \mathcal{E}_{traj}(x_{root})\|_2^2 \quad (7)$$

The final training objective for our model is a weighted sum of the primary reconstruction loss and all four auxiliary losses. The weights (λ) are hyperparameters that balance the influence of each component, ensuring that the model learns to generate motion that is simultaneously accurate, smooth, physically plausible, and spatially controlled. The total loss is defined as:

$$\begin{aligned}\mathcal{L}_{\text{total}} = \lambda_{\text{simple}}\mathcal{L}_{\text{simple}} + \lambda_{\text{joint}}\mathcal{L}_{\text{joint}} \\ + \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{traj}}\mathcal{L}_{\text{traj}}\end{aligned} \quad (8)$$

To validate our proposed method, we conducted a series of experiments, which we detail below.

IV. Experiments and Results

A. Experimental Setup

1) Dataset: Our experiments are conducted on the AIST++ dataset [Ruilong Li et al. 2021], a large-scale, multi-view dataset featuring 3D human dance motions synchronized with music across 10 distinct dance genres. We adhere to the official train/test splits for all experiments.

To further analyze the model's robustness and behavior under data-constrained conditions, we also create a filtered subset containing only motions from the Jazz Ballet (JB) genre. This stylistically homogeneous but smaller dataset allows us to stress-test the model's tendency to overfit and understand its sensitivity to the properties of the input data.

2) Preprocessing: For our training pipeline, we follow the standard data processing procedure established by prior works. The 60 FPS motion data is first downsampled to 30 FPS to match the temporal resolution of our model. Subsequently, the long motion and music sequences are sliced into fixed-size 5-second clips (150 frames), which serve as the fundamental training samples for our model. This ensures that all inputs to the network have a consistent sequence length, as required by the Transformer architecture.

3) Evaluation Metrics:

Mean Trajectory Error (MTE): To quantitatively evaluate the model's ability to precisely follow a specified path, we introduce a new metric: the Mean Trajectory Error (MTE). This metric measures the average deviation between the generated motion's path and the ground truth control trajectory. For each generated sequence in the test set, we first align the predicted and ground truth sequences to the same length. We then compute the Euclidean distance (L2 norm) between the 3D position of the predicted root joint and the 3D position of the ground truth root joint at every corresponding frame. The MTE for a single sequence is the average of these distances over all frames. The final reported MTE is the mean of these values across all sequences in the test set. Lower values indicating higher accuracy and better adherence to the control trajectory.

$$\text{MTE} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_{\text{pred}}^{(i)} - \mathbf{p}_{\text{gt}}^{(i)} \right\|_2, \quad (9)$$

where N is the total number of frames in the sequence, $\mathbf{p}_{\text{pred}}^{(i)}$ is the predicted 3D root position vector at frame i, $\mathbf{p}_{\text{gt}}^{(i)}$ is the ground truth 3D root position vector at frame i, and $(|\cdot|_2)$ is the Euclidean distance (L2 norm).

Physical Foot Contact (PFC) Score: To evaluate physical plausibility, we adopt the Physical Foot Contact (PFC) score

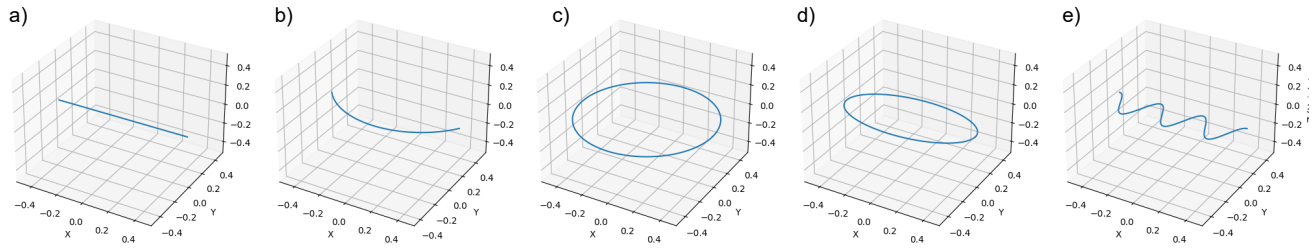


Fig. 3. **T**rajectory Types: Simplest: a) line and b) semicircle; Simpler: Simplest + c) circle and d) ellipse; Complex: Simpler + e) sinusoidal

from EDGE [Tseng, Castellon, and Liu 2023]. This metric is designed to quantify the "foot skate" artifact by penalizing horizontal foot velocity during periods of high root acceleration when feet should be static. The score is calculated as a product of these values at each frame and then averaged over the sequence. As an error metric, lower PFC scores indicate more realistic motion with less foot skating.

4) Implementation Details: Our proposed model builds upon the EDGE architecture, integrating an additional transformer encoder to process trajectory information. The baseline model was retrained for 1700 epochs, while our models were trained for 1500 epochs due to computational constraints. We investigate two key design choices:

- **Trajectory Generation Dynamics**: We analyze the effect of the trajectory's temporal dynamics using two distinct generation methods for our test set. The first, **Low-Dynamism (Full-then-Sliced)**, involves segmenting a single, continuous path (a line or semicircle) generated over the entire song duration, which creates predictable, low-velocity conditioning signals. The second, **High-Dynamism (Per-Slice)**, generates a complete, dynamic shape independently within each 5-second slice. These high-dynamism trajectories were tested in three configurations of increasing geometric complexity: Simplest (lines, semicircles), Simpler (adding circles and ellipses), and Complex (adding sinusoidal curves), Fig. 3.
- **Trajectory Normalization**: For our ablation studies, we compare two standard normalization techniques applied to the input trajectory data: (i) Min-Max Scaling, which scales the data to a fixed range of [-1, 1], and (ii) Z-Score Normalization, which transforms the data to have a mean of 0 and a standard deviation of 1.

B. Quantitative Results

Main Results on the Full Dataset: We first evaluate our model against the baseline on the full AIST++ test set. As shown in Table I, our trajectory-conditioned model demonstrates a dramatic improvement in trajectory adherence. Using Min-Max normalization, our model achieves an MTE of 0.5674, a 71.8% reduction compared to the baseline's MTE of 2.0096. This result strongly validates the effectiveness of our approach in conditioning the dance generation on a specified path. However, we observe an increase in the PFC score relative to the baseline (3.292 vs. 0.945), which suggests a potential trade-off between strict trajectory following and the fine-grained physical plausibility of the footwork, where the model prioritizes adherence to the global path.

To better understand the nature of this trade-off, we conducted a follow-up analysis on the impact of the trajectory's temporal dynamics, with results detailed in Table II. This experiment reveals that the low-dynamism 'Full-then-Sliced' trajectories achieve the best Physical Foot Contact (PFC) score of 3.29, indicating the most physically plausible motion. However, this comes at the cost of the highest Mean Trajectory Error (MTE) at 0.56 among the tested configurations. Conversely, conditioning on high-dynamism trajectories, particularly the 'Simplest' configuration, yields the best trajectory-following performance with an MTE of 0.43, but at the expense of a significantly higher PFC score of 5.02. This demonstrates that while the predictable, low-velocity signals of the sliced trajectories encourage smoother, more natural dance generation, the stronger, more varied signals from the per-slice trajectories enable more precise spatial control, albeit by compromising the physical realism of the resulting choreography.

TABLE I
QUANTITATIVE RESULTS ON THE FULL AIST++ DATASET. LOWER IS BETTER FOR BOTH METRICS. BEST RESULTS ARE IN **BOLD**.

Model	Normalization	PFC (↓)	MTE (↓)
Baseline	-	**0.945**	2.0096
Ours	Min-Max	3.292	**0.5674**

TABLE II
QUANTITATIVE RESULTS ON THE FULL AIST++ DATASET USING MIN-MAX NORMALIZATION WITH THE INPUT TRAJECTORY CONFIGURATIONS.

Trajectory Type	PFC Score (↓)	MTE (↓)
Complex	6.38	0.47
Simpler	6.51	0.51
Simplest	5.02	0.43
Full→Sliced	3.29	0.56

C. Ablation Studies

To better understand our model's sensitivities, we performed a targeted ablation study on the JB subset. This study analyzes the impact of both the trajectory's dynamic properties and the chosen normalization scheme on the model's learning process.

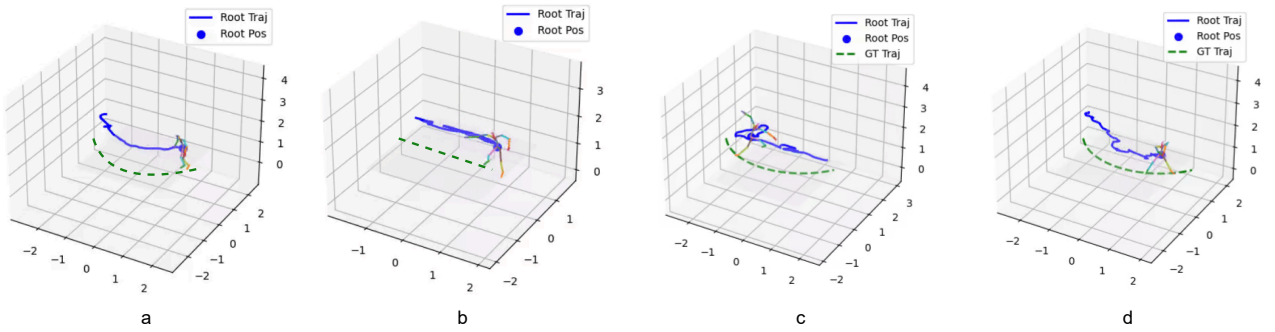


Fig. 4. **Q**ualitative Results: a) and b) show results on full dataset, the generated dances follow trajectory; c) JB Dataset Min-Max Normalization, and d) JB Dataset Z-Normalization

1) Impact of Trajectory Dynamics on Mode Collapse: Our initial experiments with low-dynamism (Full-then-Sliced) trajectories revealed a critical failure mode. As shown in Table III, the model achieves exceptionally low error scores. However, our qualitative analysis confirmed that this was due to motion collapse: the model learned a trivial solution of generating a static pose. This minimizes MTE against a slow-moving target and yields an artificially low PFC due to the complete lack of movement. To counteract this, we introduced high-dynamism (Per-Slice) trajectories. This more challenging conditioning signal successfully prevented complete motion collapse. However, as shown in Table IV, it also revealed a trade-off between trajectory adherence and choreographic quality, with different normalization schemes pushing the model towards different behaviors. (Note: Simplest Trajectory configuration was used for High-Dynamism Results.)

2) Analysis of Normalization Scheme Trade-offs: The results with high-dynamism trajectories Table IV highlight a fundamental tension in the model when trained on the smaller JB dataset. The two normalization schemes produced distinct behaviors:

- **Z-Score Normalization** led the model to heavily prioritize the trajectory objective. It achieved a very low MTE of 0.4320, indicating excellent path following. However, this came at the cost of choreographic variety. Qualitatively, the model generated a mostly static or frozen pose (e.g., a single ballet stance) that would "glide" along the target path. This can be seen as a form of pose collapse, where the model sacrifices dance motion to perfectly satisfy the spatial constraint. The relatively low PFC score (3.084) is a symptom of this limited limb movement.
- **Min-Max Scaling** produced the opposite outcome. The model failed to effectively learn the trajectory conditioning, resulting in a very high MTE of 1.6433. However, it continued to generate dynamic, varied dance poses in response to the music. This uncontrolled but active motion led to a very high PFC score of 8.414, indicating significant foot-skating as the model "danced" without adhering to its spatial target.

Ultimately, these findings highlight the critical role that both the nature of the conditioning signal and the specifics of data normalization play in guiding the model's behavior, especially under data-constrained conditions.

TABLE III
ABLATION RESULTS ON THE JB DATASET USING LOW-DYNAMISM (FULL-THEN-SLICED) TRAJECTORIES. THE DECEPTIVELY LOW SCORES ARE A RESULT OF MODEL OVERFITTING AND MOTION COLLAPSE.

Normalization	PFC (↓)	MTE (↓)
Min-Max	**0.311**	0.6264
Z-Norm	2.954	**0.1976**

TABLE IV
ABLATION RESULTS ON THE JB DATASET USING SIMPLEST HIGH-DYNAMISM (PER-SLICE) TRAJECTORIES. THESE TRAJECTORIES MITIGATE MOTION COLLAPSE, FORCING THE MODEL TO GENERATE DYNAMIC MOVEMENT.

Normalization	PFC (↓)	MTE (↓)
Min-Max	8.414	1.6433
Z-Norm	**3.084**	**0.4320**

D. Qualitative Results

Our qualitative assessments of rendered animations strongly support our quantitative findings. On the full dataset, our model successfully generates dances that follow the global trajectory, as illustrated in Figure 4 (a) and (b), where the dancer's path clearly traces the target semicircle and line, respectively.

The ablation study on the JB dataset provides critical visual insights. The high-dynamism trajectories produce two different outcomes. With Min-Max scaling (Fig. 4(c)), the model generates a sequence of varied dance poses, but its global path deviates from the target trajectory. With Z-Normalization (Fig. 4(d)), the model's root position follows the path, but the body remains in a nearly-frozen pose, effectively "gliding" through space. This visual evidence confirms the trade-off between spatial adherence and choreographic quality discovered in our quantitative analysis.

V. Discussion

Our experimental results demonstrate that while trajectory-conditioned dance generation is feasible, it introduces a delicate balance between spatial control and choreographic quality. This is most apparent when the model is trained on a data-scarce, stylistically-focused domain like the Jazz Ballet subset. On this data, there is a clear trade-off: Z-Score normalization excels at spatial accuracy but can sacrifice dance quality via pose collapse, while Min-Max scaling preserves dance quality but fails on spatial accuracy.

On the large and diverse AIST++ dataset, the model finds a more effective equilibrium. It successfully learns to follow user-defined paths as evidenced by a 71.8% reduction in MTE compared to the baseline without suffering the catastrophic pose or motion collapse observed on the JB subset. Nonetheless, the trade-off persists, as seen by the increase in the PFC score, indicating that forcing adherence to a strict path can disrupt the fine-grained physical plausibility of the generated dance. Additionally, while Z-score normalization of the trajectory led to pose collapse on the smaller dataset, this was likely due to the limited motion diversity. We hypothesize that the rich variety of motions in the full AIST++ dataset would act as a natural regularizer, preventing the model from converging on trivial solutions. Therefore, training our full model with Z-score normalization could potentially yield a superior Mean Trajectory Error (MTE), achieving even greater spatial accuracy without the stylistic degradation observed in our initial, smaller-scale experiments.

Furthermore, our analysis reveals the profound impact of the conditioning signal's temporal characteristics. Low-dynamism trajectories, while seemingly simpler, can inadvertently encourage the model to find trivial, static solutions. High-dynamism signals are crucial for preventing this failure mode and forcing the generation of genuine movement. These findings suggest that the design of the conditioning signal is as important as the model architecture itself. For this task to be successful in interactive applications, the model must be robust not only to the geometry of the path but also to its dynamic properties.

VI. Conclusion and Future Work

A. Conclusion

In this work, we introduced *RhythmRoute*, a novel framework for controllable human dance generation that successfully integrates high-level musical conditioning with low-level spatial trajectory control.

Our primary contribution is the reformulation of the diffusion training objective to incorporate the trajectory as a direct condition and a new, explicit loss term. We extended the model's core Transformer architecture to accept a target root trajectory, allowing it to learn the complex, full-body dynamics that result from following a specific path. The introduction of a dedicated trajectory loss, which works in synergy with a suite of existing physical and temporal losses, ensures that the final generated motion is simultaneously realistic, smooth, and accurate to the user's spatial command. We also introduced a new trajectory error metric designed to evaluate the accuracy of path-following in generated motions. This approach provides a powerful and intuitive new modality for artists, animators, and researchers, bridging the gap between stylistic inspiration and explicit spatial control.

B. Future Work

While our method successfully demonstrates the feasibility of dual-modality control, it also opens up several exciting avenues for future research. Our current work focuses on controlling the root trajectory. A natural and challenging extension would be to allow for simultaneous trajectory control over multiple end-effectors (e.g., hands, feet) in a dance context. This would enable the generation of complex character-environment interactions, like dancing around props or tracing patterns with the hands while following a separate path on the floor.

Another important avenue for future research is Conflict Resolution. We currently assume that the stylistic implications of the music and the geometric constraints of the trajectory do not conflict. However, a user could provide a slow, legato music piece with a sharp, zig-zagging trajectory. A more advanced model could be designed to handle such conflicts by: (1) automatically detecting incongruity between the conditioning signals; and (2) learning a policy to either blend the two intentions or prioritize one over the other based on a user-defined preference or a learned heuristic. This would lead to more robust and intuitive user control in unconstrained creative scenarios.

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